CSWeb 8.1.0 on cPanel Shared Hosting

Installation and Operation Guide

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About this package

This package installs **CSWeb 8.1** (CSWeb Community edition) on a shared hosting account managed with **cPanel**. It is ready to upload: no SSH, Composer or command-line tools are required to install it.

On shared hosting, CSWeb is used for **data collection and synchronization**:

- deploying CSPro applications to tablets and phones,
- synchronizing cases between CSEntry and the server,
- monitoring progress (sync report, map),
- managing interviewers and supervisors,
- downloading the data for processing with CSPro (Data Viewer, export to Stata, SPSS, R or CSV).

The **breakout** feature (automatic export of cases to relational tables) and scheduled jobs are not available on shared hosting. They require a dedicated server. Ignore the **Breakout Health** menu and the breakout options under **Settings**.

Tested configuration

The package was installed and tested in an environment reproducing a cPanel shared hosting plan:

Component	Version / setting
PHP	8.3, with `exec`, `proc_open`, `shell_exec`, `system`, `popen` disabled
Web server	Apache 2.4 with `mod_rewrite` and `.htaccess`
Database	**MariaDB 10.6** (also compatible with MySQL 8)
Limits	1 CPU core, 20 database connections per user

Results: installation, application deployment, CSEntry synchronization (including **5,000 cases in a single sync**), data download, user management and Arabic text all work. **100 to 200 devices synchronizing within the same minute completed with no error.**

Before you start

Hosting checklist

Requirement	Where to check in cPanel
PHP **8.3** (8.1 minimum)	*MultiPHP Manager*, or *Select PHP Version* on CloudLinux hosts
PHP extensions: `pdo_mysql`, `zip`, `curl`, `openssl`, `fileinfo`, `dom`	*Select PHP Version* or provider
MariaDB 10.6+ or MySQL 8.0+	*MySQL Databases*
Apache `mod_rewrite` and `.htaccess`	Provider (enabled on standard cPanel)
SSL certificate for the subdomain	*SSL/TLS Status* (AutoSSL)
Disk space: about 200 MB, plus data	*Disk Usage*
Files (inodes): about 18,000	Provider

Ask your provider one question

> *What is the maximum number of simultaneous connections per database user (`max_user_connections`) on our plan?*

A value of **20 or more** is suitable. Each synchronization uses one connection for a fraction of a second.

What you need

- The package `csweb-<version>-cpanel.zip` and its checksum file `.sha256`.
- A subdomain name, for example `csweb.your-domain.org`. This guide uses this example.
- A strong administrator password (12 characters or more).

Choose the address once. The address `https://csweb.your-domain.org` is written into every CSEntry application you deploy. Changing it during fieldwork means redeploying the application to every tablet.

Installation

Step 1. Create the subdomain

1. In cPanel, open **Domains** (or **Subdomains** on older cPanel) and create `csweb.your-domain.org`.
2. cPanel fills the **Document Root** field automatically (for example `csweb.your-domain.org` or `public_html/csweb`). Replace it with **`csweb`**, so the folder is `/home/<cpanel-user>/csweb`. The `<cpanel-user>` is the user name shown in the right column of the cPanel home page.

   Do not use a folder inside `public_html`.

Step 2. Create the database

1. Open **MySQL Databases**.
2. Create a database, for example `csweb`. cPanel adds your account prefix: note the **full name**, for example `myaccount_csweb`.
3. Create a user, for example `myaccount_cswebuser`, with a strong password.
4. Under **Add User To Database**, select the user and the database, then grant **ALL PRIVILEGES**.

Step 3. Upload and extract the package

1. Open **File Manager** and go to your **home folder** (`/home/<cpanel-user>`, the top of the folder tree), **not** inside `csweb` or `public_html`.
2. Click **Upload** and select `csweb-<version>-cpanel.zip`.
3. Back in File Manager, right-click the zip and choose **Extract**, then confirm `/home/<cpanel-user>`. The files go into the folder `csweb` created in Step 1.
4. Open `csweb`. You must see the folders `api`, `setup`, `vendor` directly inside it. If you see another `csweb` folder inside `csweb`, the zip was extracted in the wrong place: delete everything and repeat from item 1.
5. Click **Settings** (top right) and tick **Show Hidden Files (dotfiles)**. Check that the file **`.htaccess`** is present in `csweb`. Without it, the home page shows *403 Forbidden* and private files can be downloaded.
6. Delete the zip file.

> **Optional: check the download.** Compare the SHA-256 of the zip with the `.sha256` file before uploading (on Windows: `certutil -hashfile file.zip SHA256`).

Folder permissions are already correct: on cPanel, PHP runs as your account, so **do not set 777 permissions**.

Step 4. Select the PHP version and settings

Do this **after** Step 3: cPanel stores these settings in `csweb/.htaccess`, and extracting the zip replaces that file.

1. Open **MultiPHP Manager**, tick `csweb.your-domain.org` and choose **PHP 8.3** (on CloudLinux hosts: **Select PHP Version**).
2. Open **MultiPHP INI Editor**, select the subdomain and set:

Setting	Value
`memory_limit`	`256M`
`upload_max_filesize`	`64M`
`post_max_size`	`64M`
`max_execution_time`	`300`

If you extract the zip again later, repeat this step.

Step 5. Enable HTTPS

1. Open **SSL/TLS Status**, select `csweb.your-domain.org` and click **Run AutoSSL**.
2. Wait until `https://csweb.your-domain.org` opens without a certificate warning. A page from the CSWeb setup is expected at this stage.

Complete this step **before** Step 6: the wizard checks the address over HTTPS, and this address is the one written into the CSEntry applications.

Step 6. Run the setup wizard

1. Open `https://csweb.your-domain.org/setup/` (with `https://`). All requirements must show a green check. If one is red, see *Troubleshooting*.
2. Click **Next** and fill in the form:

Field in the wizard	Value
Database name	Full name from Step 2, for example `myaccount_csweb`
Hostname	`localhost`
Database username	Full user name from Step 2, with the prefix, for example `myaccount_cswebuser`
Database password	Password from Step 2
CSWeb admin password	Your strong administrator password. The user name is **`admin`**.
Timezone	Your local timezone, for example `Asia/Baghdad` or `Africa/Dakar`
Maximum execution time	`300` (pre-filled)
Path to files directory	`/home/<cpanel-user>/csweb/files` (pre-filled, keep it)
CSWeb API URL	`https://csweb.your-domain.org/api/` (pre-filled: check it starts with `https://`; the final slash is added automatically)

3. Click **Next**. The wizard creates the database and shows *Setup Complete!*.

After installation, the wizard **locks itself** (`/setup/` returns *403 Forbidden*). This is expected and protects your installation.

Step 7. Validate the installation

Complete this checklist before using the server for fieldwork.

#	Test	Expected result
1	Log in at `https://csweb.your-domain.org` as `admin`	The dashboard opens, with styled tables and menus
2	**Users**: create an interviewer account	The account appears in the list
3	From **CSPro Designer**, deploy an application to `https://csweb.your-domain.org/api`	The application appears under **Apps**
4	On a device, in **CSEntry**, add the application from the server, log in as the interviewer	The application installs
5	Enter 2 or 3 test cases and **synchronize**, preferably over mobile data	Sync completes; cases appear in **Sync Report**
6	Synchronize a second device	It downloads the shared files and data as designed
7	Download the data with **CSPro Data Viewer** (download from CSWeb, using the server address above)	All test cases are present

Then delete the test data from the dashboard's list of dictionaries (*Delete data*) and the test accounts, so that fieldwork starts from an empty database.

Operation

Interviewer accounts

- Create **one account per interviewer**. The server accepts at most **5 logins per minute for the same account from the same network**; more are refused for one minute (*Too many requests*). An account shared by several tablets fails at synchronization time.
- User names may contain **only letters and digits** (for example `int001`, not `int_001`).
- To create many accounts, use **Users > Import** with a CSV file: `username, first name, last name, role, password, email, phone`, for example `int001,Ahmed,Ali,Standard User,Pass!2026a,,`. In Excel, save as **CSV UTF-8** (required for Arabic names) and tick **CSV file has a header row** in the import window.
- Keep the `admin` password private. Create supervisor accounts for daily monitoring.
- Always use the `https://` address in CSPro Designer and CSEntry.

During fieldwork

- **Do not** change the subdomain, switch between `http://` and `https://`, delete `src/config.php`, re-run the setup or extract a new zip over the installation. Each of these stops synchronization on every tablet.
- **Do not** use *Delete data* or delete a dictionary once fieldwork has started.
- If a synchronization fails at the end of the day, the interviewer waits one minute and synchronizes again. Data stays on the tablet until it is sent.

Backups

Use the **cPanel backups**. The CSWeb **Backup** page cannot run on shared hosting: *Backup now* shows an error, and a schedule enabled on that page never runs.

- **Backup** (or **JetBackup**) in cPanel saves the database and files daily on most plans. Check the frequency and retention with your provider.
- During fieldwork, download a copy at least weekly: **Backup > Download a MySQL Database Backup** (your CSWeb database) and **Download a Home Directory Backup**.
- Data is also kept on the devices until it is synchronized. Do not delete CSEntry data on devices until it has been downloaded and checked.

Capacity

Situation	Behaviour
100 to 200 devices synchronizing within the same minute	Tested with no error
Many devices at the same second (for example 100 within 10 seconds)	A few syncs fail and must be retried
Devices sharing one network (office Wi-Fi, mobile operator NAT)	Supported, up to about 60 logins per minute from one public address (each interviewer has their own account)
Large syncs (5,000 cases at once)	Tested: about 2 seconds

For a census or a survey with several hundred interviewers synchronizing at the same time, a dedicated server is recommended.

Where to find logs

In the dashboard, **Logs**. The files are in `/home/<cpanel-user>/csweb/var/logs/` (`api-<date>.log` for synchronization, `ui.log` for the web interface). They are not accessible from the web.

Troubleshooting

Symptom	Cause and solution
A requirement is red in the setup wizard	PHP version or extension: see Step 4 and the hosting checklist. For “directories are writeable”, re-extract the zip with the File Manager.
Page shows *Composer detected issues in your platform … PHP version*	The subdomain does not use PHP 8.1 or later: repeat Step 4.
Home page or `/setup/` shows *403 Forbidden* or *Index of /* before setup	`csweb/.htaccess` is missing, or the zip was extracted into `csweb/csweb`. Repeat Step 3, then Step 4.
Wizard: *Access denied for user … (using password: YES)*	Wrong database user name or password. Use the full names with prefix from Step 2; reset the password in *MySQL Databases* if needed.
Wizard: *Access denied for user … to database …*	Database name without the account prefix, or user not added to the database (Step 2, item 4).
Wizard: *Failed to contact API server*	HTTPS is not ready or the API URL is wrong. Fix it (Step 5), correct the **CSWeb API URL** field and click **Next** again.
Pages display without style; buttons and tables do not work	Incomplete upload. Delete the `csweb` folder and extract the full zip again.
Too many requests (429)	More than 5 logins in one minute with the same account from the same network: wait one minute. Do not share accounts between tablets.
Some syncs fail at peak times; the log shows `max_user_connections`	The plan’s connection limit is reached. Ask the provider to raise it, or schedule teams to sync at different times.
Large syncs fail or time out	Check Step 4 settings (`max_execution_time`, upload limits). If the provider’s firewall (ModSecurity) blocks requests, ask them to allow `/api/` on the CSWeb subdomain.
User import: *Malformed UTF-8 characters* or *Username must contain only letters and numbers; Line number: 1*	Save the file as **CSV UTF-8** and tick **CSV file has a header row**.
Backup now shows an error	Expected on shared hosting: use the cPanel backups.
Settings, Backup or Logs page shows an error	These pages are prepared automatically on the first visit after installation: reload the page. If the error persists, see `var/logs/ui.log`.
`/setup/` returns *403 Forbidden*	Normal after installation.

Updating CSWeb

Do not update during fieldwork.

1. Make a cPanel backup (database and home directory).
2. Create the folder `/home/<cpanel-user>/new` and extract the new zip there. This creates `new/csweb`.
3. Copy into `new/csweb`, from the current installation: the file `src/config.php` (written by the setup wizard) and the folders

`files/` and `files_csweb/`.

4. Rename `csweb` to `csweb-old`, move `new/csweb` to `/home/<cpanel-user>/csweb`, and empty the folder `csweb/var/cache`.
5. Repeat Step 4 (PHP version and settings).
6. Run the checklist of Step 7. Keep `csweb-old` until the update is validated.

Support

CSWeb Community: https://github.com/BOUNADRAME/csweb-community. Report issues under *Issues* with the content of `BUILD-INFO.txt` and the relevant lines of `var/logs/`.

Installation assistance and dedicated server set-up are available from StatPro Consulting.